

Xueyan ZHANG

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Education

- Master in Mathematics, **Università di Padova**, Italy Oct 2022 – Ongoing
- **Field of Study:** Mathematics, with a focus on differential geometry
 - **Thesis:** Frobenius Manifolds and Flat Pencils of Metrics
 - **Supervisor:** Prof. Paolo Rossi
 - **Relevant Coursework:** Differential Geometry, Commutative Algebra, Calculus of Variations, Representation Theory of Groups
 - **Final Semester Grade:** 30 L/30 | **Final Year Grade:** 27.5/30 | **Overall Grade:** 24.4/30
- Bachelor of Science, **Southern University of Science and Technology (SUSTech)** Sept 2018 – July 2022
“Double First-Class Construction”, China
- **Field of Study:** Mathematics and Applied Mathematics, with a focus on measure theory
 - **Thesis:** An Example of Hausdorff Dimension
 - **Supervisor:** Prof. Bochen Liu

Selected Theoretical Research Experience

- Frobenius Manifolds and Flat Pencils of Metrics** (Master Thesis) Dec 2024 – June 2025
- **Objectives:** To establish the equivalence between Frobenius manifolds and flat pencils of metrics; to construct Frobenius structures and study their associated KdV hierarchies, with examples of Hurwitz spaces; to conduct original research to expand the research to supermanifolds.
 - **Main Work:** Conducted a theoretical investigation using tools from Riemannian geometry and complex analysis. Explored the geometric framework underlying Frobenius manifolds and their role in integrable systems.
 - **Result:** Developed a comprehensive framework linking flat pencils of metrics with Frobenius manifold structures. Currently applying this to concrete examples to identify corresponding bihamiltonian hierarchies.
- Seminar: Distinguished Triangles and Triangulated Categories** Apr 2025
- **Objectives:** To introduce the concept of triangulated categories from a categorical perspective.
 - **Main Work:** Studied definition, axioms and propositions of distinguished triangles and triangulated categories, highlighting the categorical foundations; prepared and delivered an expository talk.
 - **Result:** Delivered a compact 60-minute seminar clarifying the foundational structures of triangulated categories; received positive feedback for clear exposition.
- An Introductory Study on Well-Posedness for Moreau’s Sweeping Process** Sept 2023
- **Objectives:** To prove the existence, uniqueness, and stability of solutions for Moreau’s sweeping process in a Hilbert space.
 - **Main Work:** Modelled the process with a Lipschitz moving convex set, constructed approximate solutions via the catching-up algorithm, and proved their convergence to a unique Lipschitz continuous solution.
 - **Result:** Established well-posedness with unique solutions continuously depending on initial data. Presented the proof clearly and effectively.

Selected Projects

- Research of Pendulums with Specific Characteristics** (Mathematica) Sept 2024
- Studied pendulums with oscillating suspension points, periodic length variation, and external forcing. Derived period shift maps, plotted parameter spaces, analyzed fixed points, and computed Lyapunov indicators to interpret dynamical behavior.
- Manpower Scheduling Optimization for Skyscraper Construction** (AMPL) Jan 2024
- Formulated the problem as a linear optimization model to minimize total labor costs while satisfying construction constraints. Implemented and solved the model using AMPL, analyzing the impact of different shift arrangements.

- Achieved a score of 9.8/10.

Crime Analysis in Washington D.C. (Python)

June 2021

- Conducted data exploration and preprocessing in Python; analyzed correlations between crime patterns and variables such as time and location using visualization tools. Applied and evaluated a kNN algorithm for both crime analysis and housing price prediction.

Interpolation and Error Analysis (MATLAB)

Jan 2021

- Built and tested interpolation methods; studied error behavior;
- Visualized approximation quality and convergence using MATLAB.

Work Experience

Data Analyst (Intern), Tencent – Shenzhen, China

July 2021 – Aug 2021

- Developed Python scripts and web crawlers to automate data collection and cleaning processes.
- Contributed optimization ideas for user interface and webpage functionality to enhance user experience.
- Streamlined workflows, reducing manual effort and improving team efficiency.

Personal Math Tutor, Individual

Sept 2018 – Jan 2023

- Provided on-demand academic assistance in mathematics, addressing a wide range of questions from secondary school topics to university-level calculus, linear algebra, operations research, etc.
- Supported students from various disciplines by explaining key mathematical concepts clearly and efficiently.
- Assisted 20+ university/college students and 15+ high school students through individual support.

Scholarships and Academic Competitions

- Team Leader, China International College Students' Innovation Competition 2025 (CICSIC)
Project name: From Geometry to Physics: Frobenius Structures, Supersymmetry, and Beyond
- Team Member, China International College Students' Innovation Competition 2024 (CICSIC)
Project name: 3D Human Pose and Shape Estimation from RGB Images
- Scholarship for Outstanding Freshmen, SUSTech, 2018

Extracurricular Practical Activities

Volunteer, CKOU South China Kendo Competition

June 2021

Publicity Director and Vice President, Esports Club, SUSTech

Oct 2018 – June 2022

Skills & Interests

Language Skills

- Chinese (native), English (B2-C1), Italian (basic user), Japanese (basic user)

Computer skills

- **Programming & Tools:** Python, C++, Java, SageMath, Mathematica
- **Modeling & Optimization:** MATLAB, AMPL
- **Other Software:** $\text{\LaTeX}$, Adobe Photoshop, Adobe Premiere

Interests

- Photography, Kendo, Guzheng, Bass guitar, Rubber stamp sculpture, Rubik's cube, bird watching